

THIRANEX VIRTUAL INTERNSHIP

PROJECT REPORT

“C++ Programming Virtual Internship”

SUBMITTED BY:

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Program: B.tech

Branch: B.tech(Computer Science and Engineering)

Semester: 4

Academic Year: 2024-2028

SUBMITTED TO:

IILM University, Greater Noida

Internship Details:

Thiranex Student ID:THX-AUG2626-106

APAAR ID: 769421300256

Internship Duration: 4 Weeks(AUG 2026 - SEPT 2026)

Supported by: Thiranex

Under the Guidance of

Project Manager

Thiranex

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr./Ms. IKSHITA SINGH** , a student of **IILM University, Greater Noida**, has successfully completed **8-Week (Aug 2026 - Sept 2026)** Thiranex Virtual Internship Program.

During the internship, he/she worked on the project titled “ **C++ Programming**” . His/Her performance during the internship was found to be satisfactory, and he/she demonstrated good technical and professional skills.

We wish him/her success in future endeavours.

Sagarabala Behera

Mentor, Thiranex

Date: 25-08-2026

DECLARATION

I hereby declare that the work presented in this Online Virtual Internship Project Report is my original work carried out under the guidance of my industry mentor and faculty guide. This work has not been submitted earlier for the award of any degree or diploma.

IKSHITA SINGH

(Signature)

Date : 25-08-2026

ACKNOWLEDGEMENT

I express my sincere gratitude to **Thiranex** for providing me with the opportunity to undertake this online virtual internship. I would like to thank my industry mentor **Mr./Ms. Sagarabala Behera** for his/her valuable guidance, continuous support, and technical insights throughout the internship period. I am also thankful to the Department of B.Tech(Computer Science and Engineering) for their encouragement and support

ABSTRACT

The **C++ Programming Internship at Thiranex** provided practical exposure to programming concepts and project-based software development.

C++ is a powerful programming language widely used for developing efficient and performance-oriented applications. During the internship, the focus was on strengthening programming fundamentals and applying them to practical problems.

The internship provided an opportunity to work in a remote, project-based environment under industry mentorship. Through practical activities and project-oriented learning, the internship helped in improving logical thinking, problem-solving skills, programming ability, and understanding of software development practices.

The experience also provided exposure to writing structured and maintainable C++ programs and understanding how programming concepts can be applied to develop solutions for real-world requirements.

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1. INTRODUCTION

Internships provide students with an opportunity to apply academic knowledge to practical situations and understand industry-oriented development practices.

The **C++ Programming Internship at Thiranex** was undertaken as a project-based internship from **26 August 2026 to 25 September 2026**.

The internship was conducted remotely and focused on practical programming activities under industry mentorship. The program provided an opportunity to strengthen knowledge of C++ programming and develop a structured approach toward solving programming problems.

Through project-based activities, programming concepts can be applied to develop efficient and logical solutions. The internship also helps students understand the importance of code organization, debugging, testing, and documentation in software development.

2. ORGANIZATION PROFILE

EduSkills

Organisation: Thiranex

Domain: Thiranex Skill Development & Future Work

Program: Thiranex Virtual Internship

Thiranex provides students with opportunities to gain industry-relevant knowledge and practical exposure through virtual internship programs.

The Thiranex Virtual Internship program is designed to bridge the gap between academic education and industry requirements by providing students with structured learning experiences.

Thiranex

The C++ Programming Virtual Internship completed during this period was **Thiranex**.

Thiranex is a global technology company operating across multiple technology and industrial domains. Its support for the internship provided students with exposure to industry-oriented learning and the importance of data-driven technologies in modern organisations.

3. INTERNSHIP OVERVIEW

- Internship Mode: **Online / Virtual**
- Duration: **4 Weeks**
- Platform Used: **Thiranex**
- Role: **Intern**

4. PROBLEM STATEMENT & OBJECTIVES

Problem Statement

Programming students often learn C++ concepts theoretically but need practical experience to apply those concepts to real-world programming problems.

The internship focused on bridging this gap by providing a **project-based environment** where C++ programming concepts could be applied to practical problems.

The objective was to develop programming solutions using structured C++ code while improving logical reasoning, problem-solving ability, debugging skills, and software development practices.

Objectives

The major objectives of the internship were:

- To strengthen C++ programming fundamentals.
- To improve logical and computational thinking.
- To apply Object-Oriented Programming concepts.

5. PROJECT DESCRIPTION & METHODOLOGY

The internship followed a practical and project-oriented methodology.

Step 1 – Understanding Requirements

The first step involved understanding the requirements of the assigned programming tasks or project.

The requirements were analyzed to determine the expected input, processing, and output.

Step 2 – Designing the Solution

After understanding the requirements, an appropriate solution approach was designed.

This involved identifying:

- Required data structures
- Algorithms
- Classes and objects
- Functions
- Input/output requirements
- Program flow

Step 3 – C++ Implementation

The solution was implemented using C++.

The program was structured into appropriate functions, classes, and modules wherever required.

Step 4 – Testing

The developed programs were tested using different inputs to verify whether the expected output was obtained.

Step 5 – Debugging

Errors encountered during development were identified and corrected.

Debugging helped improve the reliability and correctness of the implementation.

Step 6 – Optimization

The implementation was reviewed to improve code quality and efficiency where necessary.

Step 7 – Documentation

The project work and learning outcomes were documented as part of the internship report.

6. TOOLS & TECHNOLOGIES USED

Programming Language – C++

C++ was the primary programming language used during the internship.

It supports procedural, object-oriented, and generic programming approaches and is suitable for developing efficient applications.

Object-Oriented Programming

Important OOP concepts include:

- Classes
- Objects
- Encapsulation
- Inheritance
- Polymorphism
- Abstraction

Data Structures

Depending on the assigned tasks, data structures such as:

- Arrays
- Strings
- Linked Lists
- Stacks
- Queues
- Trees
- Hash Tables

can be used for organising and processing data efficiently.

Algorithms

Algorithmic techniques are used to solve programming problems efficiently.

Examples include:

- Searching
- Sorting
- Traversal
- Recursion
- Basic optimisation techniques

Development Environment

A C++ development environment/IDE can be used to write, compile, execute, and debug programs.

Suggested tools:

- Visual Studio Code
- GCC / G++ Compiler
- Git
- GitHub

7. IMPLEMENTATION & RESULTS

7.1 Requirement Analysis

The assigned programming tasks were first analysed to understand their requirements and expected behaviour.

The input, processing logic, and output requirements were identified before implementation.

7.2 Program Design

The solution was divided into smaller logical components to make the implementation easier to understand and maintain.

Functions and classes were used where appropriate to organise the code.

7.3 C++ Implementation

The project/task implementation was carried out using C++.

The implementation focused on:

- Correct program logic
- Proper use of C++ syntax
- Efficient data handling
- Modular code structure
- Appropriate use of functions
- Object-oriented design where required

7.4 Testing

Testing was performed using different test cases.

The testing process helped verify:

- Correct input handling
- Expected program output
- Boundary cases
- Invalid input scenarios
- Logical correctness

7.5 Debugging

During development, programming errors were identified and resolved through debugging.

Common categories of errors that can occur during C++ development include:

- Syntax errors
- Logical errors
- Runtime errors
- Incorrect input handling
- Memory-related issues

Debugging helped improve the reliability of the implementation.

8. LEARNING OUTCOMES

The internship provided the following learning outcomes:

Technical Skills

- Improved C++ programming skills.
- Strengthened understanding of programming fundamentals.

Problem-Solving Skills

- Improved logical thinking.
- Learned to break complex problems into smaller tasks.

Professional Skills

- Gained experience working in a remote environment.
- Understood project-based development.

9. CHALLENGES FACED & SOLUTIONS

Challenge 1 – Understanding Complex Requirements

Some programming tasks may initially be difficult to understand because of multiple requirements.

Solution

The problem was divided into smaller components and each requirement was analysed separately before implementation.

Challenge 2 – Debugging Errors

Errors during C++ development can prevent a program from producing the expected result.

Solution

The code was analysed systematically to identify syntax, logical, and runtime errors. Individual sections were tested to isolate the source of the problem.

Challenge 3 – Choosing the Correct Data Structure

Different problems require different methods of storing and accessing data.

Solution

The requirements and operations needed by the program were analysed before selecting the appropriate data structure.

Challenge 4 – Writing Efficient Code

A program may work correctly but still contain unnecessary operations.

Solution

10. CONCLUSION

The **C++ Programming Internship at Thiranex** provided valuable practical exposure to project-based programming and software development.

During the internship, the focus was on applying programming concepts to practical problems and improving logical reasoning, coding, debugging, and testing skills.

The remote/project-based nature of the internship provided an opportunity to understand how programming tasks can be approached systematically under industry mentorship.

Overall, the internship helped strengthen C++ programming fundamentals and provided a foundation for further learning in software development, Data Structures and Algorithms, and application development.

11. FUTURE SCOPE

The knowledge and skills developed during the internship can be further extended into advanced areas of software development.

Future learning can include:

- Advanced Data Structures and Algorithms
- Competitive Programming
- Advanced Object-Oriented Programming
- STL in C++
- System Programming
- Database Applications
- C++ GUI Development
- Backend Development
- Game Development
- Embedded Systems
- Performance Optimisation
- Multithreaded Programming
- Software Engineering

The experience can also be used as a foundation for developing larger C++ projects and contributing to open-source software.

12. REFERENCES

The following resources may be used for further learning:

Books / Learning Resources

1. Bjarne Stroustrup, *The C++ Programming Language*.
2. Programming and Data Structures course materials.
3. C++ programming practice resources.

Online Documentation

4. C++ Standard Library documentation.
5. GCC / G++ documentation.
6. Git documentation.
7. GitHub documentation.

13. ANNEXURE

The following documents can be attached to the final internship report:

1. Internship Completion Certificate
2. GitHub Repository link : <https://github.com/Ikshita1402>

